Aayan Akram

437-518-4241 | akrama11@mcmaster.ca | LinkedIn | GitHub | Portfolio

Education

McMaster University — Hamilton, ON, Canada
B.Eng. in Computer Engineering | GPA: 3.52/4.00 — *Expected May 2029*

Experience

Research Assistant – UAV Radar Signal Processing & Machine Learning — May 2026 – Present
ETF Lab, Faculty of Engineering – McMaster University — *Hamilton, ON, Canada*

- Developed a MATLAB radar-processing and validation pipeline to reconstruct UAV trajectories from simulated detections and compare Constant-Velocity and Constant-Acceleration Kalman filters
- Built a reproducible 35-trial dataset across seven maneuver classes, engineering 29 radar- and tracker-derived temporal motion features for classification
- Trained and evaluated UAV maneuver classifiers using grouped Leave-One-Trial-Out cross-validation, achieving 80.6% accuracy and 0.81 macro-F1 with a tuned random forest

IoT Systems Developer — July 2026 – Present
Kaizen Radical — *Quebec, Canada*

- Developed ESP32-S3 firmware in C++/FreeRTOS for event-driven visual-management stations, integrating interrupt-driven digital inputs, analog controls, rotary encoders, PWM indicators, persistent state, and fault-recovery logic
- Engineered bidirectional MQTT/TLS communication with Odoo ERP, implementing offline event buffering, automatic reconnection and state synchronization, watchdog recovery, telemetry, heartbeat reporting, and OTA firmware updates for scalable multi-device deployment

Embedded Computer Vision Intern — April 2026 – July 2026
Confidential Early-Stage Venture — *Arlington, VA, USA*

- Developed and deployed custom YOLO object-detection pipelines on NVIDIA Jetson hardware using Python, PyTorch, and OpenCV, enabling real-time inference across two mobile embedded prototypes
- Built and trained custom datasets in Roboflow, iterating on model and deployment parameters for compute- and latency-constrained edge inference ahead of broader autonomy-stack integration

Projects

Autonomous Search-and-Rescue UAV | *ArduPilot, MAVLink, NVIDIA Jetson, Python, PyTorch, TensorRT, OpenCV*

- Engineered a 6S, 10-inch autonomous UAV integrating ArduPilot flight control, GPS navigation, long-range telemetry, and an NVIDIA Jetson companion computer for onboard search-and-rescue perception
- Developed and deployed a YOLO-based human-detection pipeline using PyTorch and OpenCV, optimizing inference with TensorRT for real-time edge deployment and benchmarking latency and throughput against the PyTorch baseline
- Integrated companion-computer detections with the flight stack over MAVLink/UART, transmitting validated human-detection alerts to the ground station and validating the end-to-end perception-to-operator pipeline in flight

STM32-Based Flight Controller | *STM32F405, Altium Designer, Embedded C/C++, FreeRTOS, SPI, I2C, UART*

- Designed, assembled, and brought up a custom STM32F405 flight-controller, completing schematic capture and PCB layout in Altium and integrating multi-rail power, SWD, IMU/GPS interfaces, motor outputs, and power integrity/EMI considerations
- Developed real-time C/C++ firmware and peripheral drivers using FreeRTOS, interrupts, and DMA for sensor acquisition, calibration, attitude estimation, telemetry, failsafe handling, and closed-loop PID motor control
- Validated power rails, communication buses, timing, sensor acquisition, and motor outputs with an oscilloscope and logic-analyzer, bench testing, and flight-data logging

Additional Experience

Systems Engineering Intern | *Arm, Manchester, UK* — May 2024
Accelerator Physics Intern | *STFC Daresbury Laboratory, Warrington, UK* — July 2024 – August 2024
Simulation Engineering Intern | *Siemens, Manchester, UK* — February 2024

Technical Skills

Programming & Systems: C, C++, Python, MATLAB, Simulink, Linux, FreeRTOS
Embedded & Robotics: STM32, ESP32, ESP-IDF, NVIDIA Jetson, ArduPilot, MAVLink, MQTT, SPI, I2C, UART
AI & Engineering Tools: PyTorch, TensorRT, OpenCV, YOLOv8, Altium Designer, LTspice, Autodesk Inventor, SOLIDWORKS